

Ariad

Chrome Extension for Intelligent Web Content Organization

Version: 1.0

Product Type: Browser Extension

Platform: Google Chrome

Status: Concept / Prototype

1. Product Overview

1.1 Product Description

Ariad is a Chrome extension designed to help users organize, preserve, and rediscover important web content. The tool automatically structures browser tabs into meaningful categories based on their content and context, reducing the effort required to manually manage large numbers of open tabs.

Instead of relying on traditional browser history, Ariad creates an intelligent knowledge structure that allows users to quickly find previously visited websites through an intuitive search interface.

The extension supports users in managing information overload by transforming unstructured browser activity into an organized and searchable collection of relevant web resources.

2. Problem Statement

Modern users frequently interact with large amounts of online information. During research, studying, or professional work, users often open dozens of browser tabs to collect relevant resources.

However, this creates several problems:

- Important websites become difficult to find among many unrelated tabs.
- Users forget where specific information was discovered.
- Browser history only provides chronological information without context.
- Manually creating bookmarks or folders requires additional effort.
- Closing tabs can result in losing valuable information.

Current browser solutions mainly focus on storing URLs but do not understand the meaning or purpose behind visited content.

Ariad addresses this problem by automatically organizing and structuring web information.

3. Product Vision

Vision Statement

To create a smarter way of managing digital information by transforming browser activity into an organized, searchable personal knowledge space.

4. Product Goals

The main goals of Ariad are:

Primary Goals

- Reduce cognitive overload caused by excessive browser tabs.
- Automatically organize web content without manual effort.
- Enable users to quickly rediscover previously visited information.
- Provide a more meaningful alternative to traditional browser history.

Secondary Goals

- Improve productivity during research workflows.
 - Reduce time spent searching for lost information.
 - Support long-term knowledge preservation.
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5. Target Users

Persona 1: Students & Researchers

Description

Students often collect many online sources during projects, assignments, and exam preparation.

Needs

- Quickly access previous research material.
- Keep different topics separated.
- Avoid losing important resources.

Pain Points

- Hundreds of open tabs.
- Difficult to remember where information was found.
- Browser history is inefficient.



Persona 2: Knowledge Workers

Description

Professionals working with large amounts of digital information.

Examples:

- Developers
- Designers
- Researchers
- Content creators

Needs

- Organize project-related websites.
- Switch efficiently between tasks.
- Maintain access to previous information.



Pain Points

- Information scattered across different browser sessions.
- Time wasted searching

6. User Stories

Tab Organization

As a user, I want my browser tabs to be automatically categorized, so that I can understand and manage my current information context.

Search

As a user, I want to search previously visited websites by keywords, so that I can quickly rediscover relevant information.

Restoration

As a user, I want to restore previously closed tabs, so that I can continue unfinished tasks.

Information Management

As a user, I want websites to be stored with additional context, so that I can understand why they were important.

7. Functional Requirements

7.1 Automatic Tab Categorization

Description

The system automatically analyzes browser tabs and assigns them to meaningful categories.

Requirements

The system should:

- Detect currently open tabs.
- Extract relevant metadata.
- Analyze website information.
- Assign appropriate categories.

Example Categories

- Education
- Work
- Finance
- Shopping
- Entertainment
- Technology

7.2 Metadata Collection

Each stored website should contain relevant information.

Example:

Attribute	Description
Title	Website title
URL	Website address
Domain	Website provider

Attribute	Description
Category	Assigned group
Timestamp	Time visited
Last Access	Last interaction
Frequency	Number of visits

7.3 Search Functionality

Users can search stored content.

Search should support:

- Website title
- URL
- Domain
- Category

Example:

User searches:

"Java recursion"

System displays:

- Java recursion tutorial
- Programming documentation
- Previous learning resources

7.4 Tab Restoration

Users can restore previously visited websites.

Requirements:

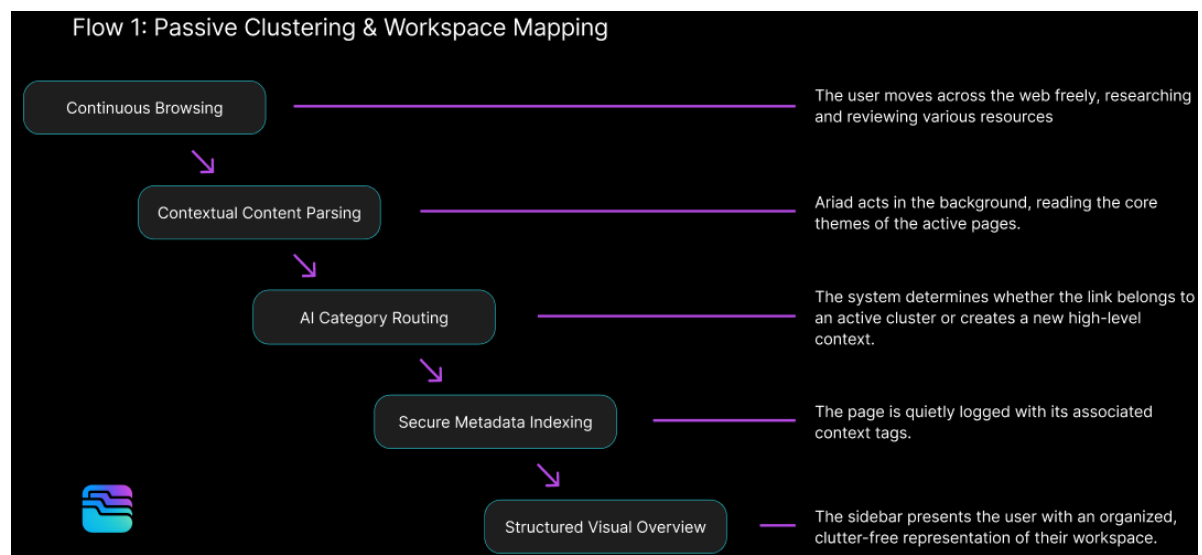
- Reopen closed tabs.
- Restore previous sessions.
- Maintain original URLs.

7.5 Category Management

Users can:

- View categories.
- Open grouped tabs.

8. User Flow



9. Technical Requirements

Platform

Google Chrome Extension

Architecture

Manifest Version 3

Technologies

Frontend:

- HTML
- CSS
- JavaScript

Browser APIs:

- Chrome Tabs API
 - Chrome Storage API
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10. Limitations

Current prototype limitations:

- Limited AI understanding.
 - Category accuracy depends on available metadata.
 - Browser-specific implementation.
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11. Future Improvements

Possible future developments:

- Cross-device synchronization.
 - Personalized recommendations.
 - Browser history integration.
 - Collaborative knowledge spaces.
 - Advanced AI assistant.
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12. Conclusion

Ariad provides a new approach to browser information management by transforming passive browsing history into an active, structured knowledge system. Through automatic categorization, intelligent search, and easy restoration, the extension helps users reduce information overload and efficiently rediscover valuable web content.